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PAPUA NEW GUINEA

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FIRST REVIEWS UNDER EXTENDED ARRANGEMENT UNDER THE EXTENDED FUND FACILITY AND AN ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY, AND REQUEST FOR MODIFICATION OF QUANTITATIVE PERFORMANCE CRITERIA —DEBT SUSTAINABILITY ANALYSIS

Approved By
Sanjaya Panth (IMF, APD) and Jarkko Turunen (IMF, SPR), Manuela Francisco and Lalita Moorty (both IDA)

Prepared by the staff of the International Monetary Fund (IMF) and the International Development Association (IDA)

Papua New Guinea: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*No*

Papua New Guinea (PNG) remains at high risk of debt distress under the Low-Income Country Debt Sustainability Framework (LIC DSF), with weak debt-carrying capacity.[1, 2] *While the planned fiscal consolidation helps address debt vulnerabilities exacerbated by the global COVID-19 shock, the risk of both external and public debt distress continues to be assessed as high. Over the medium-term, public debt enters a downward trend and the projected temporary breaches of sustainability indicators can mostly be addressed by debt management operations as well as improvements in revenue generation. The Debt Sustainability Analysis (DSA) suggests that PNG is susceptible to trade-related and contingent liabilities shocks, underscoring downside risks to the debt outlook in a global environment of high uncertainty. To lower the risk of debt distress and ensure debt sustainability, gradual fiscal consolidation, including by boosting revenues, and steadfast structural reforms to promote private sector growth would be needed. Conditional on the implementation of the authorities' plans for further fiscal consolidation and conservative financing strategies, PNG's external and overall debt is judged as sustainable.*

[1] This Debt Sustainability Analysis has been prepared jointly by the International Monetary Fund and the World Bank, in accordance with the revised Debt Sustainability Framework for low-income countries approved by the Executive Boards of the IMF and the International Development Association.

[2] The Composite Indicator (CI) of 2.56 is based on the latest available CI information—April 2023 IMF World Economic Outlook (WEO) and the World Bank's Country Policy and Institutional Assessment (CPIA) for 2021 indicating a "weak" capacity to carry debt.

PUBLIC SECTOR DEBT COVERAGE

1. The coverage of public debt in the DSA is unchanged from the previous (March 2023) DSA (Text Table 1). The segments of the public sector captured in the DSA include the central government, state and local government, and guarantees to other entities in the public and private sector, including parts of state-owned enterprises (SOEs). However, debt numbers do not fully capture implicit government guaranteed debts of SOEs and unfunded superannuation liabilities relating to pensions.[3] For the purposes of this DSA, the coverage of public sector debt remains unchanged from the last DSA, which was prepared in March 2023 in the context of the IMF ECF/EFF program request. Given continued difficulties in capturing and assessing SOE risks, a contingent liabilities stress test is included in this DSA, assuming 9 percent of GDP as SOE debt is not captured in official public debt data (the stock of explicit government guarantees is around 1.3 percent of GDP), and 3 percent of GDP for other elements of general government (mainly unfunded superannuation liabilities related to pensions, which are projected to be 2.1 percent of GDP in 2023). Separately, according to the World Bank's PPP database, the PPP capital stock in PNG is zero and, therefore, no default shock is triggered. A financial market shock of 5 percent is added, reflecting the average fiscal cost of financial crisis in low-income countries. With these assumptions, the cumulative shock in the contingent liabilities stress test amounts to 17 percent of GDP—compared to 7 percent under default assumptions. Currency denomination is used to define external debt.

Text Table 1. Papua New Guinea: Coverage of Public Debt

	Subsectors of the public sector	Subsection Covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The central, state, and local governments, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	3	Unfunded superannuation liabilities relating to pensions
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	9	SOE sovereign guarantee in dispute
4	PPP	35 percent of PPP stock	0	Default used
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	Default used
	Total (2+3+4+5) (in percent of GDP)		17	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

[3] Comprehensive data on the debt stock of SOEs in PNG is not publicly available. The ADB have been supporting the authorities with a three-year program (2020-2023) aiming to reform the governance of state-owned enterprises, increase their financial transparency and ensure financial sustainability. This included providing advice on the overarching policy and legal framework for SOEs, leading to the adoption of an amended Kumul Consolidated Holdings Act in 2021.

BACKGROUND ON DEBT

2. Between 2017 and 2022, the stock of public debt in PNG increased from around 24 billion Kina to more than 53 billion Kina. This was mainly due to external loans, while the creditor composition has been gradually shifting away from commercial loans towards official multilateral and bilateral financing (*Text Table 2*). However, expensive commercial loans have generally been replaced with official multilateral and bilateral financing at more favorable conditions. This has helped to improve debt sustainability indicators in the medium-term. Public and Publicly Guaranteed (PPG) external debt figures used for this DSA are consistent with the information in the World Bank's International Debt Statistics.

Text Table 2. Papua New Guinea: Decomposition of Public Debt Service by Creditors[1], 2022-2024

	Debt Stock (end of period)			Debt Service					
	2022			2022	2023	2024	2022	2023	2024
	(In US$ million)	*(Percent total debt)*	*(Percent GDP)*	*(In US$ million)*			*(Percent GDP)*		
Total	14969	100	47	5340	4654	4856	17	15	15
External	7426	50	24	717	408	575	2	1	2
Multilateral creditors[2,3]	3364	22	11	385	184	262	1	1	1
IMF	709	5	2	1	6	6	0	0	0
World Bank	545	4	2	134	33	47	0	0	0
ADB	2015	13	6	244	138	202	1	0	1
Other Multilaterals	95	1	0	6	7	8	0	0	0
Bilateral Creditors[2]	3482	23	11	273	165	265	1	1	1
Paris Club	1907	13	6	181	50	136	1	0	0
o/w: Australia	1397	9	4	172	35	122	1	0	0
Non-Paris Club	1575	11	5	93	115	129	0	0	0
o/w: China EXIM	1174	8	4	68	90	97	0	0	0
Bonds	500	3	2	42	42	42	0	0	0
Commercial creditors	80	1	0	17	17	7	0	0	0
Other international creditors	0	0	0	0	0	0	0	0	0
Domestic	7543	50	24	4623	4247	4281	15	14	13
Held by residents, total	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Held by non-residents, total	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
T-Bills	3479	23	11	3778	3698	3577	12	12	11
Bonds	3216	21	10	669	434	557	2	1	2
Loans	849	6	3	176	114	147	1	0	0
Memo items:									
Collateralized debt[2,4]	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Contingent liabilities	816	5	3	n/a	n/a	n/a	n/a	n/a	n/a
Nominal GDP	31533	n/a	100						

1/As reported by Country authorities according to their classification of creditors, including by official and commercial. Debt coverage is the same as the DSA.

2/Some public debt may not be shown in the table due to confidentiality clauses/capacity constraints.

3/Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into Arrears)

4/Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt. Collateralization entails a borrower granting liens over specific existing assets or future receivables to a lender as security against repayment of the loan. Collateral is "unrelated" when it has no relationship to a project financed by the loan. An example would be borrowing to finance the budget deficit, collateralized by oil revenue receipts. See the joint IMF-World Bank note for the G20 "Collateralized Transactions: Key Considerations for Public Lenders and Borrowers" for a discussion of issues raised by collateral.

3. The IMF's general allocation of Special Drawing Rights (SDRs) became effective in August 2021, with SDR 252 million (US$ 357 million, or 95.7 percent of quota) allocated to PNG. The

authorities used the full SDR allocation to support the 2021 budget. They used the SDR allocation to replace costly financing, which has helped reduce the budget deficit. For the purpose of this DSA, the SDR allocation is included in total public debt while the associated debt service for the amount outstanding is also reflected.

4. The new 38-month IMF program, approved by the Board in March 2023 will support PNG's reform agenda, help protect the vulnerable and foster inclusive growth. These facilities total SDR 684.3 million under the IMF's Extended Credit Facility (ECF) and the Extended Fund Facility (EFF) arrangements. The program builds on progress under the past SMPs (2020/21 and 2021/22), focusing on addressing structural impediments to growth and poverty reduction. Program objectives focus on (i) strengthening debt sustainability through a multi-year fiscal consolidation program while making room to meet critical social and development needs, (ii) enhancing the legal and operational framework of the BPNG to alleviate FX shortages and transition to a market-clearing exchange rate, and (iii) build on improvements to governance and the anti-corruption framework.

5. PNG is an IDA blend country, with total IDA20 allocation at SDR 173.7 million. PNG is currently eligible to access the Regional Window, the Crisis Response Windows, and the Private Sector Window. However, PNG is not eligible for SUW-SMLS as the country has a set-aside in place due to unmet FY23 PPAs (see below). The IDA decision to graduate a country to IBRD-only status is based on an assessment of the country's macroeconomic prospects, risk of debt distress, vulnerability to shocks, institutional constraints, and levels of poverty and social indicators.

6. The IDA Sustainable Development Finance Policy (SDFP) supports PNG in addressing key debt vulnerabilities. As part of the SDFP, PNG is subject to debt-related Performance and Policy Actions (PPAs). For fiscal year 2022, these were aimed at limiting non-concessional borrowing and operationalizing the 2021 State Guarantee Policy, both of which were satisfactorily implemented. PNG had two SDFP PPAs for fiscal year 2023: (i) a US$ 1 billion PPG external borrowing limit for new non-concessional long-term contractual obligations, which applies continuously throughout FY23, and (ii) to improve management of fiscal risks by both approving a revised on-lending policy (which introduces credit risks assessments and strengthens enforcement arrangements, recording and reporting requirements) and refraining from any new on-lending until the revised policy is adopted. PNG missed both actions, so the country is subject to a set-aside of 20 percent from its FY24 IDA country allocation. Additionally, due to the breach of the non-concessional borrowing ceiling (NCB) in FY23, PNG is required to implement an additional PPA on fiscal sustainability in FY24. Recovery of the set-aside at the start of FY25 will be contingent on: (i) implementation of FY23 PPA2 (carry-over PPA), (ii) compliance with the NCB ceiling PPA; and (iii) implementation of the PPA on fiscal sustainability in FY24.

BACKGROUND ON MACROECONOMIC FORECASTS

7. Growth in 2023 would be driven by services in the non-resource sector (Text Table 3). After a strong recovery in 2022, real GDP growth is expected to remain elevated at 3.0 percent in 2023, driven by the non-resource sector (4.6 percent). Resource sector growth is forecasted to contract by 1.4 percent. In 2022, average headline inflation was significantly lower than expected at 5.3 percent (3.4 percent, year-on-

year by yearend), helped by fiscal measures introduced by the government to address the cost-of-living impact following Russia's invasion of Ukraine and the decrease in seasonal item prices including alcoholic beverages and tobacco in the last quarter of the year. Headline inflation (period-average CPI) is forecasted to moderate to 2.2 percent in 2023, as global commodity prices slow and supply chain constraints ease. Following the transition to BPM6 from BPM5 and Fund TA support (see Annex VII of the SR), the BOP data has been substantially revised leading to a lower current account balance. Consequently, for 2023, the current account balance is expected to be significantly lower, by 5.4 percentage points of GDP, at 18.6 percent relative to the March 2023 projections. This follows the downwardly revised current account surplus of 22.8 percent of GDP in 2022.[4] This is the most substantial change since the 2022 Article IV and SMP review which projected a surplus of 23 percent of GDP. The immediate direct impact of the resource sector windfall on the fiscal balance is expected to be minimal. In 2022, goods exports volumes are estimated to have increased by 6.5 percent compared to 2021, with widespread increases across mineral resource export volumes, including gold, silver, copper and nickel. In addition, buoyant commodity prices in 2022 were supportive of export values, which increased by 49 percent compared to 2021. Expected lower commodity prices in 2023 underpin assumptions for a fall in nominal export growth in 2023.

Text Table 3. Papua New Guinea: DSA Vintages: Macroeconomic and Fiscal Assumptions, 2023-2028

	DSA Vintage	2023	2024	2025	2026-2028
Real GDP growth y/y (in percent)	**2023 ECF/EFF first rev.**	**3.0**	**5.0**	**3.1**	**3.1**
	2023 ECF/EFF program req.	3.7	4.4	3.1	3.1
Resource sector	2023 ECF/EFF first rev.	-1.4	6.3	0.1	0.1
	2023 ECF/EFF program req.	0.3	5.1	0.1	0.1
Non-resource sector	2023 ECF/EFF first rev.	4.6	4.6	4.1	4.0
	2023 ECF/EFF program req.	4.9	4.2	4.2	4.1
Inflation, annual average (consumer prices, percent)	**2023 ECF/EFF first rev.**	**2.2**	**4.0**	**4.8**	**4.6**
	2023 ECF/EFF program req.	5.4	4.9	4.6	4.5
Current account balance (percent of GDP)	**2023 ECF/EFF first rev.**	**18.6**	**16.3**	**16.7**	**12.3**
	2023 ECF/EFF program req.	24.0	22.4	24.8	23.0
Growth of exports of G&S (US$, in percent)	2023 ECF/EFF first rev.	-8.0	4.8	3.6	2.9
	2023 ECF/EFF program req.	-11.2	3.1	2.3	2.7
Growth of imports of G&S (US$, in percent)	2023 ECF/EFF first rev.	-2.5	0.5	3.1	2.9
	2023 ECF/EFF program req.	3.6	1.5	2.6	3.3
Primary balance (percent of GDP)	**2023 ECF/EFF first rev.**	**-1.9**	**-1.5**	**-0.2**	**1.8**
	2023 ECF/EFF program req.	-2.2	-1.2	0.3	2.2
Government revenues (excluding grants, percent of GDP)	2023 ECF/EFF first rev.	16.4	16.4	17.1	17.9
	2023 ECF/EFF program req.	15.3	15.1	15.6	16.0

Source: PNG authorities; IMF staff estimates and projections.

8. The medium-term baseline macroeconomic forecast is broadly unchanged from the March 2023 DSA (Text Table 3). At around 3 percent, the long-term potential real growth estimate remains little

[4] Previously published current account data in PNG likely overstated the balance due to long-standing challenges in classifying large income account outflows, including external debt service payments related to resource projects, under the financial account rather than the current account. Fund technical assistance (TA) in external sector statistics aimed at correcting this misclassification, while work is ongoing on improving quality of data. In this regard, authorities aim at initiating an inter-agency working group on external sector statistics including representatives from the NSO, Treasury Department, PNG Customs Service and the BPNG to improve the consistency of external sector statistics with other official statistics by mid-2024.

changed since the March 2023 DSA.[5] Inflation is projected to reach an annual average of 4.6 percent in the medium term. As strong global demand for PNG's export goods is expected to persist over the medium-term, the current account surplus is forecast to remain strong but slightly decrease to around 12 percent of GDP, in the medium-term.

9. The medium-term baseline also assumes rapid progress on fiscal consolidation, as envisaged by the authorities. This is appropriate to strengthen debt sustainability and build a fiscal buffer. While continued primary deficits are anticipated in the short-term (Text Tables 3 and 4), the projection builds in a significant amount of fiscal consolidation and primary surpluses in the second half of the projection period, consistent with the authorities' plans to meet the requirements of the Fiscal Responsibility Act, which specifies government debt should be maintained at no more than 40 percent of GDP over the long term. To contain expenditure growth, the authorities are committed to reducing current expenditure as a share of non-resource GDP, and to enhancing expenditure efficiency through payroll management improvements, stronger expenditure controls, and more efficient cash management practices. Meanwhile, revenue mobilization is a critical component of the planned consolidation in a context of elevated development needs. In the near-term, revenue is expected to increase following the Non-Tax Revenue Act (NTRA), and the updated dividend policy, which is expected to result in higher revenue from the resource sector. Immediate revenue reform priorities include implementation of the Tax Administration Act (TAA) and introducing amendments to the Income Tax Act (ITA). From 2026 onward, the authorities also project an increase in tax revenues from the PNG LNG project as tax exemptions expire. Further, the authorities see scope for significantly higher dividends after loan amortization for the project is completed but this is not yet included in the baseline in full. Achieving a balanced budget by 2027 would lower the risks from significant debt service obligations on external borrowing coming due in 2028.

10. The main downside risks to the baseline projection include: natural disasters, lower global growth, and social or political instability. PNG is vulnerable to natural disasters (flooding, landslides and earthquakes) as well as the impact of climate change (through droughts and sea level rises), and El Ninio.[6] Lower global growth would likely impact PNG through lower commodity prices, with adverse consequences for the balance of payments and budget through lower resource revenue. Although the 2022 general elections were accompanied by several violent disruptions, political and social risks appear to have fallen since then. With limited sources of financing available in an adverse scenario, and continued pressing social and development needs, the room for significant policy adjustment is relatively limited. Engagement with the IMF through the ECF/EFF program provides an important anchor for the authorities to advance their reform agenda. If growth deteriorated significantly compared to the projections, further debt buildup may be needed to finance the budget and maintain government services. Upside risks also exist and include higher-than-expected commodity prices, or start of any of several major projects,

[5] Expected new resource projects are not included in the (authorities' or staff's) baseline, explaining the low resource sector growth in the medium-term. Higher growth in the resource sector due to new projects is therefore a major upside risk to the baseline projection.

[6] Authorities are working closely with partners to develop climate mitigation and adaptation programs through the Green Climate Fund; develop green financing standards through the BPNG; and fund the UN Climate and Biodiversity Trust Fund in our 2023 budget. Moreover, there is a continued and growing need for support in this area.

including Papua LNG, P'nyang LNG or the Wafi Golpu mining, which are not yet in the baseline scenario (see Annex III of the SR).

Text Table 4. Papua New Guinea: Summary of Fiscal Operations, 2023-2025

	2023	2024	2025
Revenue and grants (percent of GDP)	**18.2**	**18.0**	**18.5**
Taxes	14.5	14.5	15.2
Grants	1.8	1.7	1.4
Other revenue	1.8	1.8	1.8
Expenditure (percent of GDP)	**22.6**	**22.0**	**21.0**
Expense	18.5	18.2	17.4
Compensation of employees	6.1	6.2	5.9
Use of goods and services	6.6	6.5	6.3
Interest	2.5	2.5	2.3
Other	3.3	3.1	2.8
Net acquisition of nonfinancial assets	4.1	3.8	3.6
	0.0	0.0	0.0
Gross operating balance	2.2	2.3	3.4
Net lending (+)/borrowing (-)	-4.4	-3.9	-2.5
Primary balance (percent of GDP)	**-1.9**	**-1.5**	**-0.2**

Source: PNG authorities; IMF staff estimates and projections.

11. The LIC DSA's realism tools indicate that the government's primary balance adjustment is moderately ambitious. At about 3.0 percent, the three-year cumulative adjustment of the primary balance is within the top 20 percent of historical experiences, relative to peers. Much of this adjustment has already taken place during 2023 (Figure 4). Baseline growth projection is broadly in line with the implied path consistent with the range fiscal multipliers. A low base, unwinding of Covid-19 and high commodity prices support activity while the Porgera gold mine is expected to reopen in 2024, further contributing to strong growth in the resource sector.

12. Financing mix: For domestic financing, the DSA assumes that the composition of T-bills and T-bonds is broadly similar to the past six years (*i.e.*, 75 percent of new issuances corresponding to T-bills) in the first half of the forecast horizon, but T-bill financing gradually decreases over time. For the near-term, the DSA considers all existing commitments. Although the profile of domestic debt, with a high concentration in short-term Treasury bills, raises concerns over rollover risks, short-term liquidity risks and reliance on domestic financing fall throughout the projection horizon because the level of newly issued debt falls markedly as fiscal deficits are replaced by surpluses, particularly after 2027. While the trend in the projection period remains constant, continued development of the domestic debt market is necessary to increase liquidity and transition towards greater reliance upon domestic financing sources, while shifting to longer maturities.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TEST

13. PNG's debt carrying capacity is assessed as weak. According to the April 2023 World Economic Outlook and the 2021 Country Policy and Institutional Assessment (CPIA), PNG's Composite Indicator (CI) is 2.56, indicating weak debt-carrying capacity (Text Table 5).[7] Hence, the applicable thresholds are 30 percent for the present value (PV) of external debt-to-GDP ratio; 140 percent for the PV of the external debt-to-exports ratio; 10 percent for the external debt service-to-exports ratio; 14 percent for the external debt service-to-revenue ratio; and 35 percent for the PV of public debt-to GDP ratio, respectively.

Text Table 5. Papua New Guinea: Composite Indicator and Applicable Thresholds

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.851	1.10	43%
Real growth rate (in percent)	2.719	2.304	0.06	2%
Import coverage of reserves (in percent)	4.052	55.959	2.27	88%
Import coverage of reserves^2 (in percent)	-3.990	31.314	-1.25	-49%
Remittances (in percent)	2.022	0.002	0.00	0%
World economic growth (in percent)	13.520	2.856	0.39	15%
CI Score			2.56	100%
CI rating			Weak	

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintage
Weak	Weak	Weak	Weak
	2.56	2.58	2.62

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

[7] At 2.56 PNG's CI is close to the weak/medium threshold of 2.69.

14. Scenario stress tests: As indicated in the section on public debt coverage, a contingent liabilities stress test is included to account for SOE debt not captured in official public debt data. [8] Further, given the size and importance of PNG's resource sector (with a share of commodities in total exports of goods and services of 96 percent), a commodity price shock is included in the DSA. Considering the high price volatility over the past few years, the fuel price shock is set at 35 percent (compared to the default shock of 58 percent), and the shock to non-fuel commodity prices is set to 24 percent—with 20 percent for base metals and precious metals, and 31 percent for agricultural commodities other than grain (price shocks to grain like wheat, corn, and soybeans, are not relevant for PNG and, therefore, not included in the stress test). Mitigating factors are included as well, and at default values (2 percent for fuel, and 27 percent for non-fuel). PNG's single outstanding Eurobond (maturing in 2028) activates the market financing module.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

15. Under the baseline scenario, the debt-service to revenue indicator is projected to breach its threshold. A relatively large breach in 2028 arises due to the bullet payment for the US$500 million Eurobond, which was issued in 2018. Marginal breaches arise in 2026, 2027 and 2029 as the debt-service-to-revenue ratio surpasses 14, the threshold value, due to higher official bilateral and multilateral debt service payments given tighter financial conditions, which is falling by 2030. In the baseline scenario, the present value of debt-to-GDP ratio as well as debt-to-exports and debt service-to-export ratios remain below their respective thresholds over the entire projection horizon. The solvency indicators are on a downwards trend in the latter half of the projection.

16. Stress tests point to vulnerabilities in PNG's external debt dynamics particularly with respect to exports shocks, which would cause threshold breaches for all four external sustainability indicators. Changes in policy and the structure of the economy manifests in some differences between the historical and the baseline scenarios. However, with the revision of the BOP data, the historical scenario is now closer to the baseline owing to the reclassification of some current account items. The market financing risk module indicates a moderate risk of heightened liquidity pressures primarily due to elevated GFNs. However, a heightened market stress event would not have a substantial impact of debt burden indicators as few future external debt disbursements are projected on commercial terms (Figure 5). PNG's relatively elevated sovereign spreads (which have dropped by close to 150 basis points since the ECF/EFF approval) likely reflect the perceived risks due to the country's characteristics (small and undiversified export base, small revenue base, vulnerability to shocks).

17. Although residuals remain relatively large, the current DSA shows an improvement to the assessment of debt dynamics with smaller residuals (Table 1 and Figure 3). With support of Fund TA, the recently published BOP data in BPM6 addressed some issues that led to overestimating the residuals,

[8] During the 2021 SMP, the authorities initiated an SOE reform program to reduce the backlog of audited annual financial statements and to strengthen SOE oversight and improve understanding of fiscal risks. This program includes a detailed review of SOE debt and government guarantees and is also expected to improve the reporting of public debt.

such as long-standing challenges in classifying large income account outflows, including external debt service payments related to resource projects, under the financial account rather than the current account. Some data quality issues persist, which likely continue to drive positive residuals from external financial flows which persist into the projection period as a consistent accounting framework is used.

B. Public Sector Debt Sustainability Analysis

18. Public debt PV ratios have increased substantially in recent years and are currently expected to increase to 47 percent of GDP for 2023. Starting from this level means that the public debt sustainability indicator is in breach of the threshold for countries with weak debt-carrying capacity (that is, 35 percent of GDP) during the first half of the projection horizon. Under the baseline scenario, the public debt-to-GDP ratio remains broadly stable just above 52 percent of GDP until 2025, before falling, more markedly in the second half of the projection. This downward trend in the public debt-to-GDP ratio arises through stronger real GDP growth and a smaller fiscal deficit than over the past 5 years, as growth headwinds wane and gradual fiscal consolidation continues (Figure 3). The profile of domestic debt, with a high concentration in short-term Treasury bills, raises concerns over rollover risks. Diversifying the issuance structure by issuing longer term Treasury bonds would lower these risks, make debt service costs more predictable and help with financial deepening effort (Figure 2).[9] In addition, the fall in the level of short-term debt, from a peak of 14.9bn Kina in 2022 to 8.6bn Kina in 2028, is a mitigating factor for liquidity risks. The substantial residuals from unidentified debt creating flows, arising largely in 2018, are not anticipated to be repeated (Figure 3) as the government has taken several steps to improve recording and reporting of debt.[10]

19. Stress tests point to several vulnerabilities for public debt (Figure 2 and Table 4). As for external debt, the most extreme shock impacting the PV of public debt is a shock to exports. Under this shock scenario, the PV of public debt-to-GDP ratio peaks at 62.1 percent of GDP in 2027, more than 1.5 times the 35 percent threshold value and substantially above the starting level of 22.4 percent of GDP in 2023, before gradually decreasing over the next years. The tailored stress test for the combined contingent liability shock causes a deterioration in public debt sustainability which is felt most acutely through the total public debt service-to-revenue measure. This analysis suggests contingent liabilities represent an important source of vulnerabilities in PNG. The trajectory of the PV of the public debt-to-revenue ratio is impacted most by the commodity price shock, reflecting the strong reliance upon commodity exports within PNG.

RISK RATING AND VULNERABILITIES

20. PNG remains at "high" risk of external and overall debt distress. The (mechanical) overall debt distress rating is "high", owing to the multiple breaches of sustainability thresholds under the baseline

[9] Authorities are assessing available options, including debt-for-nature swap operations, with analysis still at an exploratory stage.

[10] This includes the state guarantee policy initiated during the 2021 SMP.

scenario, as discussed in the previous section. Likewise, the external debt distress rating is unchanged from previous DSAs at high risk, given breaches in debt service-to-revenue discussed before.

21. Debt service on existing loans, paired with relatively weak revenue generation, are expected to almost double the debt service-to-revenue ratio by 2025. However, as debt service reduces and revenues increase, and barring further shocks to demand growth, the indicator enters a downward trend from this peak.

22. Stress tests show that adverse shocks to exports, commodity prices and contingent liabilities constitute the main risks to public debt sustainability. Further, the historical scenario indicates that it will be challenging to reduce debt from current levels and that reforms, including those already implemented during the recent SMPs, are essential for supporting the sustainability of public finances. Market financing risks continue to be relevant, with the GFN threshold breached, pointing to moderate market financing pressures.

23. Debt dynamics are assessed as sustainable. Public debt is expected to remain broadly stable in the near-term and to enter a clear downward path over the medium-term. Also, the projected temporary breaches of sustainability indicators can be prevented by debt management operations as well as by boosting revenue generation. External debt-to-GDP and debt-to-exports ratios are below their thresholds over the entire projection horizon. Public external and overall debt is judged to be sustainable conditional on the implementation of the authorities plans for further fiscal consolidation and conservative financing strategies (such as substituting costly financing with concessional financing from multilateral and bilateral partners). This baseline sustainability assessment also relies upon higher future resource revenue as tax exemptions expire from 2026 onwards.

AUTHORITIES' VIEWS

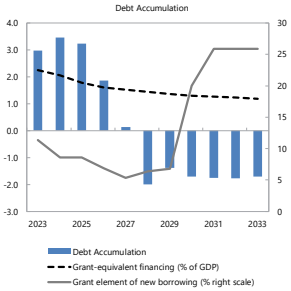
24. The authorities noted the Staff's assessment that PNG remains at high risk of debt distress but remains sustainable under the baseline projection. In discussions, the authorities agreed that large financing requirements to dampen the effects of the COVID-19 pandemic on PNG's economy have left their mark on public debt, and that redemption of the US$ 500 million Eurobond in 2028, issued in 2018, is a key risk. However, the authorities were more optimistic about their debt-carrying capacity and perceived a lower risk of debt distress, pointing to increased revenues by 2027 as debt payments for the PNG LNG project are completed. They highlighted their strategy in recent years to substitute costly financing with concessional financing from multilateral and bilateral partners, which has improved PNG's debt profile. The authorities noted interest costs of domestic securities have fallen over the past two and a half years. They also highlighted the importance of more favorable future contract negotiation and the medium-term revenue strategy as mitigating factors which will reduce risks in the medium-term. The authorities are committed to fiscal consolidation and conservative financing strategies to support the sustainability of PNG's debt going forward.

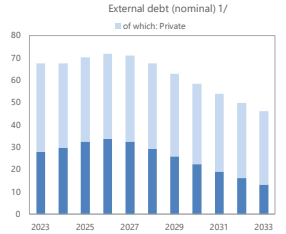
Table 1. Papua New Guinea: External Debt Sustainability Framework, Baseline Scenario, 2020-2043

(In percent of GDP, unless otherwise indicated)

	Actual			Projections																	Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	Historical	Projections
External debt (nominal) 1/	**69.0**	**68.5**	**60.7**	**67.5**	**67.5**	**70.2**	**71.7**	**70.9**	**67.4**	**46.0**	**42.7**	**39.7**	**37.3**	**35.2**	**33.3**	**31.6**	**30.0**	**27.7**	**25.6**	**23.8**	**77.2**	**62.4**
of which: public and publicly guaranteed (PPG)	**21.8**	**24.8**	**23.6**	**27.8**	**29.7**	**32.4**	**33.6**	**32.5**	**29.2**	**13.3**	**10.9**	**9.0**	**7.5**	**6.3**	**5.3**	**4.4**	**3.6**	**2.9**	**2.3**	**1.9**	**13.9**	**25.6**
Change in external debt	7.3	-0.5	-7.8	6.8	0.0	2.7	1.5	-0.8	-3.4	-3.8	-3.3	-2.9	-2.4	-2.1	-1.9	-1.7	-1.6	-2.3	-2.1	-1.9		
Identified net debt-creating flows	**-15.4**	**-26.1**	**-39.8**	**-22.2**	**-21.9**	**-21.1**	**-19.3**	**-16.3**	**-14.5**	**-12.8**	**-12.6**	**-12.5**	**-12.4**	**-12.4**	**-12.4**	**-12.3**	**-12.3**	**-12.4**	**-12.4**	**-12.4**	**-15.7**	**-16.6**
Non-interest current account deficit	**-17.4**	**-16.3**	**-25.2**	**-19.8**	**-17.5**	**-17.8**	**-15.9**	**-14.5**	**-12.8**	**-11.3**	**-11.1**	**-10.8**	**-10.5**	**-10.3**	**-10.1**	**-9.9**	**-9.6**	**-9.4**	**-9.2**	**-9.0**	**-14.3**	**-14.4**
Deficit in balance of goods and services	-16.0	-16.3	-26.4	-24.3	-24.6	-25.1	-25.5	-25.6	-25.4	-21.9	-21.3	-20.6	-20.0	-19.4	-18.9	-18.3	-17.8	-17.2	-16.7	-16.2	**-14.2**	**-24.3**
Exports	38.5	43.5	46.1	44.4	43.4	44.2	44.9	45.0	44.8	38.8	37.8	36.7	35.7	34.7	33.8	32.9	32.0	31.1	30.3	29.5		
Imports	22.4	27.2	19.7	20.1	18.8	19.1	19.3	19.4	19.4	16.9	16.5	16.1	15.7	15.3	14.9	14.6	14.2	13.9	13.6	13.3		
Net current transfers (negative = inflow)	-1.1	-1.1	-0.8	-1.0	-0.9	-0.9	-0.9	-0.9	-0.8	-0.6	-0.6	-0.5	-0.5	-0.5	-0.4	-0.4	-0.4	-0.4	-0.3	-0.3	**-1.0**	**-0.8**
of which: official	-1.4	-1.7	-1.3	-1.3	-1.2	-1.2	-1.2	-1.1	-1.1	-0.8	-0.8	-0.7	-0.7	-0.6	-0.6	-0.6	-0.5	-0.5	-0.5	-0.4		
Other current account flows (negative = net inflow)	-0.2	1.1	2.0	5.5	8.0	8.1	10.5	11.9	13.4	11.2	10.8	10.4	10.0	9.6	9.2	8.9	8.5	8.2	7.9	7.6	**0.9**	**10.7**
Net FDI (negative = inflow)	**-3.7**	**-6.5**	**-5.7**	**-1.6**	**-2.4**	**-2.3**	**-2.3**	**-2.3**	**-2.2**	**-1.8**	**-1.9**	**-2.1**	**-2.2**	**-2.4**	**-2.6**	**-2.8**	**-3.0**	**-3.2**	**-3.4**	**-3.6**	**-2.4**	**-2.0**
Endogenous debt dynamics 2/	**5.7**	**-3.3**	**-8.9**	**-0.7**	**-2.0**	**-0.9**	**-1.0**	**0.5**	**0.5**	**0.4**	**0.4**	**0.3**	**0.3**	**0.3**	**0.3**	**0.3**	**0.3**	**0.2**	**0.2**	**0.2**		
Contribution from nominal interest rate	3.4	3.1	2.4	1.2	1.1	1.1	1.1	2.6	2.6	1.8	1.7	1.6	1.5	1.4	1.3	1.3	1.2	1.1	1.0	1.0		
Contribution from real GDP growth	2.0	0.0	-2.5	-1.9	-3.2	-2.1	-2.2	-2.1	-2.1	-1.5	-1.3	-1.3	-1.2	-1.1	-1.0	-1.0	-0.9	-0.9	-0.8	-0.8		
Contribution from price and exchange rate changes	0.3	-6.4	-8.9	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
Residual 3/	**22.7**	**25.6**	**32.0**	**28.9**	**21.9**	**23.8**	**20.8**	**15.5**	**11.1**	**9.0**	**9.3**	**9.6**	**10.0**	**10.3**	**10.5**	**10.6**	**10.7**	**10.1**	**10.3**	**10.5**	**14.8**	**15.2**
of which: exceptional financing	-1.6	0.0	0.0	-0.6	-0.8	-0.9	-0.3	0.2	0.2	0.3	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators																						
PV of PPG external debt-to-GDP ratio	**...**	**...**	**18.4**	**22.4**	**24.2**	**26.9**	**28.4**	**27.9**	**25.0**	**11.4**	**9.3**	**7.6**	**6.3**	**5.2**	**4.3**	**3.5**	**2.8**	**2.2**	**1.7**	**1.2**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**40.0**	**50.5**	**55.6**	**60.9**	**63.3**	**61.9**	**55.8**	**29.3**	**24.6**	**20.6**	**17.5**	**15.0**	**12.7**	**10.7**	**8.7**	**7.0**	**5.5**	**4.2**		
PPG debt service-to-exports ratio	**12.7**	**8.1**	**5.3**	**3.1**	**3.8**	**4.9**	**5.9**	**6.3**	**9.6**	**6.0**	**5.2**	**4.5**	**3.4**	**2.9**	**2.5**	**2.3**	**2.1**	**1.9**	**1.6**	**1.4**		
PPG debt service-to-revenue ratio	**37.8**	**27.6**	**15.8**	**8.5**	**10.0**	**12.6**	**15.3**	**15.7**	**23.7**	**11.5**	**9.9**	**8.4**	**6.2**	**5.1**	**4.4**	**3.9**	**3.5**	**3.2**	**2.6**	**2.1**		
Gross external financing need (Million of U.S. dollars)	-2228.1	-3230.2	-7126.1	-3994.9	-4137.9	-4137.1	-4243.4	-2713.0	-1707.9	-3096.4	-3512.1	-3963.2	-4538.6	-5041.4	-5542.7	-6052.1	-6600.3	-7199.9	-7891.7	-8621.7		
Key macroeconomic assumptions																						
Real GDP growth (in percent)	-3.2	0.1	4.3	3.0	5.0	3.1	3.1	3.0	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	**3.8**	**3.3**
GDP deflator in US dollar terms (change in percent)	-0.5	10.3	14.9	-7.3	2.1	-1.2	-1.8	-0.7	0.5	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1	**0.6**	**0.7**
Effective interest rate (percent) 4/	5.2	5.0	4.2	1.8	1.8	1.7	1.6	3.7	3.8	3.9	3.9	4.0	4.0	4.0	4.1	4.1	4.1	4.0	4.0	4.0	**4.8**	**3.0**
Growth of exports of G&S (US dollar terms, in percent)	-18.4	24.8	26.9	-8.0	4.8	3.6	2.9	2.7	3.1	3.4	3.4	3.4	3.4	3.5	3.5	3.5	3.5	3.5	3.6	3.6	**10.2**	**2.4**
Growth of imports of G&S (US dollar terms, in percent)	-14.3	34.0	-13.3	-2.5	0.5	3.1	2.7	2.8	3.4	3.6	3.6	3.7	3.7	3.8	3.8	3.8	3.9	3.9	3.9	4.0	**-1.9**	**2.5**
Grant element of new public sector borrowing (in percent)	...	...	...	11.3	8.6	8.6	6.9	5.3	6.4	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	25.9	**...**	**13.8**
Government revenues (excluding grants, in percent of GDP)	12.9	12.8	15.4	16.4	16.4	17.1	17.4	18.0	18.2	20.1	20.0	19.8	19.7	19.6	19.5	19.4	19.2	19.1	19.0	18.9	**15.4**	**18.4**
Aid flows (in Million of US dollars) 5/	1736.2	1411.5	1043.1	727.3	678.5	556.8	526.6	531.2	523.0	571.7	589.4	605.2	624.0	642.2	662.2	682.2	703.5	725.1	747.7	770.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	2.3	2.0	1.8	1.6	1.5	1.4	1.2	1.2	1.1	1.1	1.0	1.0	1.0	1.0	0.9	0.9	0.9	**...**	**1.5**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	38.9	34.9	30.3	33.7	50.1	59.4	98.6	98.6	99.3	99.4	99.7	99.7	99.9	99.9	99.9	99.9	100.0	**...**	**66.0**
Nominal GDP (Million of US dollars)	23,848	26,312	31,533	30,104	32,268	32,869	33,273	34,044	35,269	48,010	51,065	54,313	57,769	61,444	65,353	69,511	73,933	78,637	83,640	88,961		
Nominal dollar GDP growth	-3.6	10.3	19.8	-4.5	7.2	1.9	1.2	2.3	3.6	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4	**4.3**	**4.0**
Memorandum items:																						
PV of external debt 7/	...	...	55.6	62.1	61.9	64.7	66.5	66.3	63.2	44.1	41.0	38.3	36.1	34.1	32.3	30.7	29.1	26.9	24.9	23.1		
In percent of exports	...	...	120.6	139.8	142.6	146.4	148.2	147.1	140.9	113.5	108.6	104.3	101.0	98.2	95.7	93.3	91.1	86.5	82.3	78.3		
Total external debt service-to-exports ratio	30.6	24.1	17.9	18.4	16.2	17.2	12.2	19.6	22.7	17.2	16.2	15.1	13.8	13.0	12.4	11.9	11.5	11.0	10.4	9.8		
PV of PPG external debt (in Million of US dollars)			5814.7	6753.2	7796.0	8837.6	9450.4	9496.1	8814.4	5459.5	4747.6	4102.7	3616.0	3195.0	2809.3	2437.8	2066.9	1702.7	1383.9	1100.7		
(PVt-PVt-1)/GDPt-1 (in percent)				3.0	3.5	3.2	1.9	0.1	-2.0	-1.7	-1.5	-1.3	-0.9	-0.7	-0.6	-0.6	-0.5	-0.5	-0.4	-0.3		
Non-interest current account deficit that stabilizes debt ratio	-24.7	-15.8	-17.4	-26.6	-17.5	-20.6	-17.4	-13.7	-9.4	-7.5	-7.7	-7.9	-8.1	-8.2	-8.2	-8.1	-8.0	-7.1	-7.1	-7.1		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

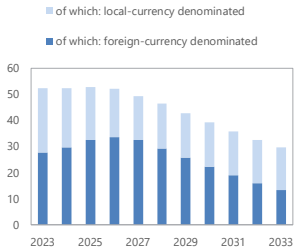
Table 2. Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020-2043

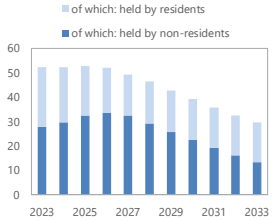
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2033	2043	Historical	Projections
Public sector debt 1/	**48.7**	**52.2**	**48.4**	**52.4**	**52.2**	**52.7**	**52.1**	**49.2**	**46.5**	**29.6**	**14.1**	**37.4**	**44.1**
of which: external debt	**21.8**	**24.8**	**23.6**	**27.8**	**29.7**	**32.4**	**33.6**	**32.5**	**29.2**	**13.3**	**1.9**	**13.9**	**25.6**
Change in public sector debt	8.1	3.5	-3.8	4.0	-1.2	0.5	-0.6	-2.8		-2.9	-0.7		
Identified debt-creating flows	**9.7**	**1.6**	**-3.7**	**0.9**	**-1.3**	**-1.0**	**-1.8**	**-3.2**	**-2.9**	**-2.4**	**-0.7**	**3.2**	**-2.1**
Primary deficit	**6.2**	**4.4**	**3.0**	**1.9**	**1.5**	**0.2**	**-0.9**	**-2.5**	**-2.1**	**-2.0**	**-0.5**	**3.2**	**-1.1**
Revenue and grants	14.7	15.0	16.7	18.2	18.0	18.5	18.8	19.4	19.6	21.3	19.7	**17.2**	**19.8**
of which: grants	1.7	2.3	1.3	1.8	1.7	1.4	1.4	1.4	1.4	1.2	0.9		
Primary (noninterest) expenditure	20.9	19.4	19.7	20.1	19.5	18.7	17.8	16.9	17.6	19.3	19.2	**20.4**	**18.7**
Automatic debt dynamics	**3.5**	**-2.7**	**-6.6**	**-1.0**	**-2.8**	**-1.2**	**-0.9**	**-0.7**	**-0.8**	**-0.4**	**-0.1**		
Contribution from interest rate/growth differential	3.0	-1.2	-5.0	-1.0	-2.8	-1.2	-0.9	-0.7	-0.8	-0.4	-0.1		
of which: contribution from average real interest rate	1.7	-1.2	-2.8	0.4	-0.3	0.4	0.7	0.8	0.7	0.6	0.3		
of which: contribution from real GDP growth	1.3	0.0	-2.2	-1.4	-2.5	-1.6	-1.6	-1.5	-1.5	-1.0	-0.4		
Contribution from real exchange rate depreciation	0.5	-1.5	-1.6	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-1.7**	**1.9**	**-0.1**	**3.1**	**1.2**	**1.5**	**1.2**	**0.4**	**0.2**	**-0.5**	**-0.1**	**-0.3**	**0.4**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**43.3**	**47.0**	**47.2**	**47.7**	**47.4**	**44.9**	**42.6**	**27.8**	**13.4**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**259.0**	**258.7**	**261.4**	**258.6**	**252.4**	**231.5**	**217.3**	**130.8**	**68.2**		
Debt service-to-revenue and grants ratio 3/	**133.8**	**122.3**	**101.4**	**85.1**	**83.4**	**83.1**	**79.5**	**69.9**	**68.0**	**40.0**	**27.5**		
Gross financing need 4/	25.8	22.7	19.9	17.4	16.5	15.5	14.0	11.0	11.2	6.5	4.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	-3.2	0.1	4.3	3.0	5.0	3.1	3.1	3.0	3.1	3.1	3.1	**3.8**	**3.3**
Average nominal interest rate on external debt (in percent)	11.5	10.7	8.0	2.0	2.6	2.8	3.1	3.3	3.4	3.5	3.6	**18.7**	**3.1**
Average real interest rate on domestic debt (in percent)	-0.4	-9.2	-11.6	3.3	-1.5	0.5	1.7	2.0	1.1	1.8	2.2	**-0.6**	**1.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	2.6	-6.5	-6.8	...	...	...	...	...	...	...	...	**2.4**	**...**
Inflation rate (GDP deflator, in percent)	1.6	11.8	15.2	-1.8	4.2	2.9	1.9	2.2	3.1	3.1	3.1	**5.8**	**2.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	11.1	-7.2	5.9	5.4	1.9	-1.4	-1.4	-2.2	6.9	3.1	3.1	**3.5**	**3.2**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-1.9	0.9	6.8	-2.1	2.7	-0.3	-0.3	0.4	-2.1	0.9	0.2	**1.9**	**0.4**
PV of contingent liabilities (not included in public sector debt)	0.2	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

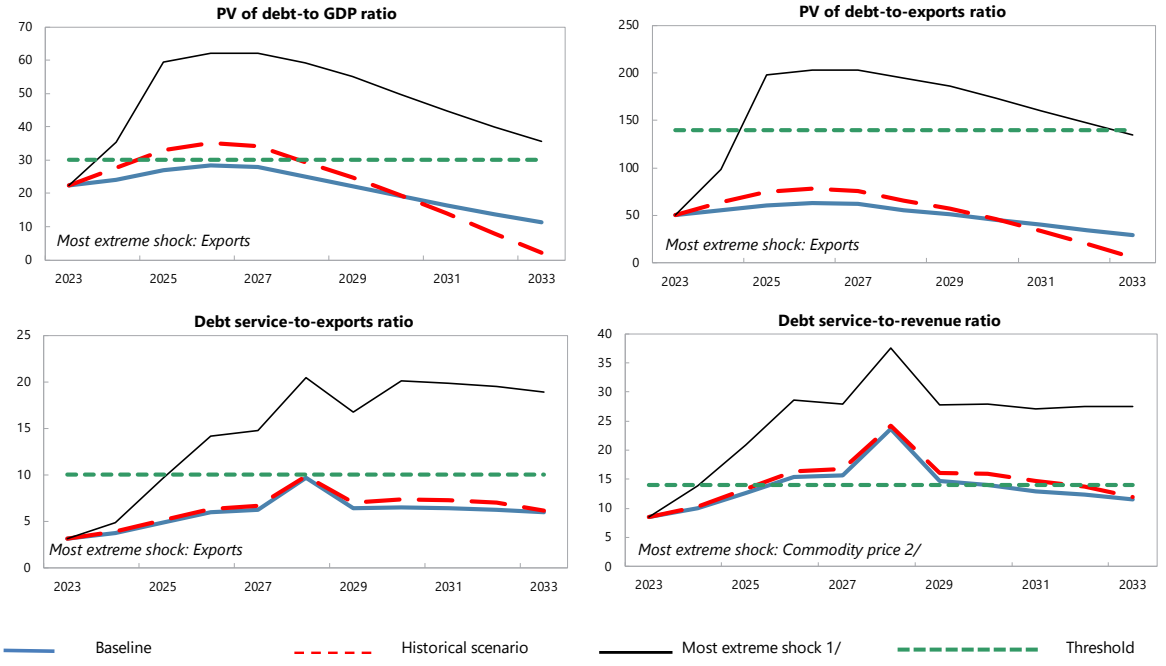
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2023-2033



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	Yes	No
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

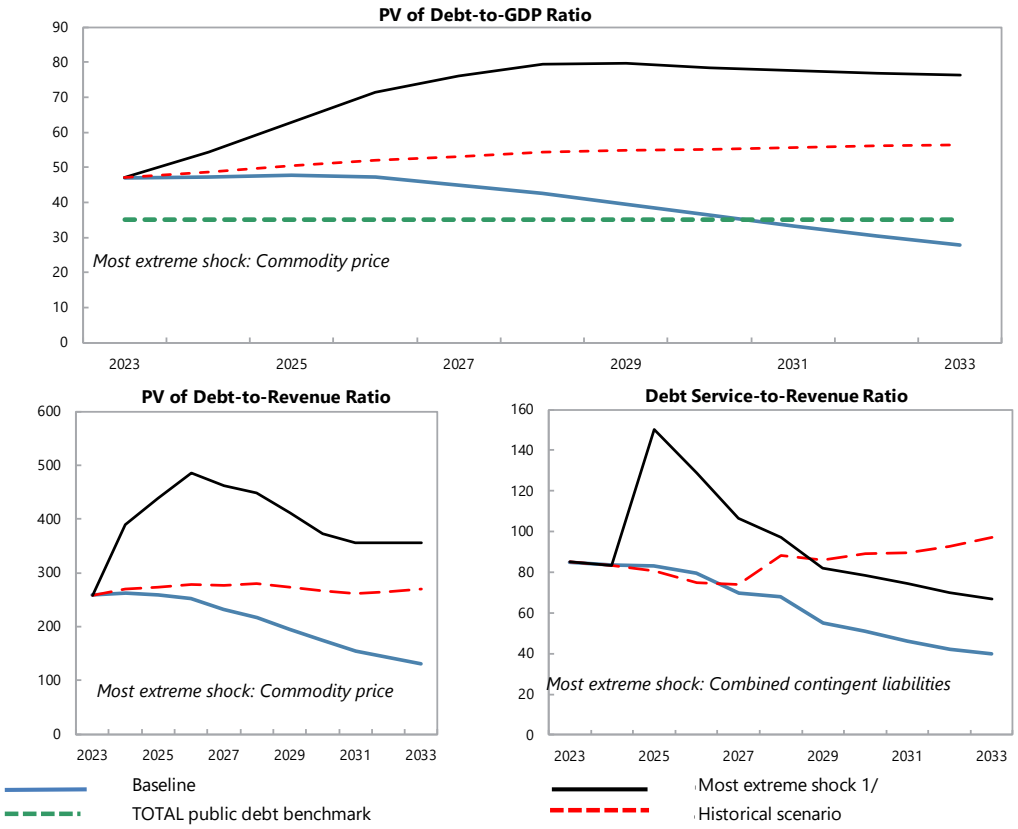
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	4.3%	4.3%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	22	22
Avg. grace period	4	4

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2023-2033



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	12%	12%
Domestic medium and long-term	15%	15%
Domestic short-term	73%	73%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	4.3%	4.3%
Avg. maturity (incl. grace period)	22	22
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	3.7%	3.7%
Avg. maturity (incl. grace period)	7	7
Avg. grace period	4	4
Domestic short-term debt		
Avg. real interest rate	1.0%	1.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2033. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2023-2033

(In percent)

	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of debt-to GDP ratio											
Baseline	22	24	27	28	28	25	22	19	16	14	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	22	28	**33**	**35**	**34**	29	25	19	14	8	2
B. Bound Tests											
B1. Real GDP growth	22	26	**31**	**33**	**32**	29	26	22	19	16	13
B2. Primary balance	22	25	29	**31**	**31**	28	25	23	20	17	15
B3. Exports	22	**35**	**59**	**62**	**62**	**59**	**55**	**50**	**45**	**40**	**36**
B4. Other flows 3/	22	27	**32**	**34**	**34**	**31**	28	24	21	18	16
B5. Depreciation	22	**30**	**37**	**40**	**39**	**35**	**32**	28	24	21	17
B6. Combination of B1-B5	22	**33**	**40**	**42**	**42**	**39**	**35**	**31**	28	24	21
C. Tailored Tests											
C1. Combined contingent liabilities	22	26	**30**	**33**	**33**	**30**	28	25	23	20	18
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	22	**34**	**47**	**53**	**57**	**58**	**58**	**56**	**55**	**53**	**51**
C4. Market Financing	22	27	30	**32**	**31**	28	25	21	18	15	13
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	51	56	61	63	62	56	51	46	40	35	29
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	51	64	75	78	76	65	57	46	34	20	6
B. Bound Tests											
B1. Real GDP growth	51	56	61	63	62	56	51	46	40	35	29
B2. Primary balance	51	57	65	69	68	63	58	53	48	43	38
B3. Exports	51	98	**198**	**203**	**203**	**194**	**186**	**173**	**160**	**147**	135
B4. Other flows 3/	51	62	73	76	75	69	64	58	52	46	40
B5. Depreciation	51	56	67	70	69	63	58	52	46	41	35
B6. Combination of B1-B5	51	78	81	107	105	98	91	84	76	68	60
C. Tailored Tests											
C1. Combined contingent liabilities	51	60	68	72	73	68	64	60	55	51	47
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	51	96	125	134	139	138	137	137	137	137	136
C4. Market Financing	51	56	61	63	62	56	51	46	40	35	29
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	3	4	5	6	6	10	6	6	6	6	6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	3	4	5	6	7	10	7	7	7	7	6
B. Bound Tests											
B1. Real GDP growth	3	4	5	6	6	10	6	6	6	6	6
B2. Primary balance	3	4	5	6	7	10	7	7	7	7	7
B3. Exports	3	5	10	**14**	**15**	**20**	**17**	**20**	**20**	**20**	**19**
B4. Other flows 3/	3	4	5	6	7	**10**	7	8	8	7	7
B5. Depreciation	3	4	5	6	7	10	7	7	7	7	7
B6. Combination of B1-B5	3	4	7	9	9	**13**	**10**	**11**	**11**	**10**	10
C. Tailored Tests											
C1. Combined contingent liabilities	3	4	5	6	7	**10**	7	7	7	7	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	3	5	7	10	**10**	**14**	**12**	**13**	**14**	**14**	**15**
C4. Market Financing	3	4	5	6	6	10	6	6	6	6	6
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	8	10	13	**15**	**16**	**24**	**15**	14	13	12	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	8	10	13	**16**	**17**	**24**	**16**	**16**	**15**	14	12
B. Bound Tests											
B1. Real GDP growth	8	11	**15**	**18**	**18**	**28**	**17**	**16**	**15**	**14**	13
B2. Primary balance	8	10	13	**16**	**16**	**25**	**16**	**15**	**14**	14	13
B3. Exports	8	11	**17**	**25**	**25**	**34**	**26**	**29**	**27**	**26**	**25**
B4. Other flows 3/	8	10	13	**17**	**17**	**25**	**17**	**17**	**15**	**15**	14
B5. Depreciation	8	13	**16**	**20**	**21**	**31**	**19**	**19**	**18**	**17**	**16**
B6. Combination of B1-B5	8	11	**16**	**20**	**20**	**29**	**21**	**20**	**19**	**18**	**17**
C. Tailored Tests											
C1. Combined contingent liabilities	8	10	13	**16**	**17**	**25**	**16**	**15**	**14**	14	13
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	8	14	**21**	**29**	**28**	**38**	**28**	**28**	**27**	**28**	**27**
C4. Market Financing	8	10	13	**15**	**16**	**24**	**15**	14	13	12	12
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt, 2023-2033

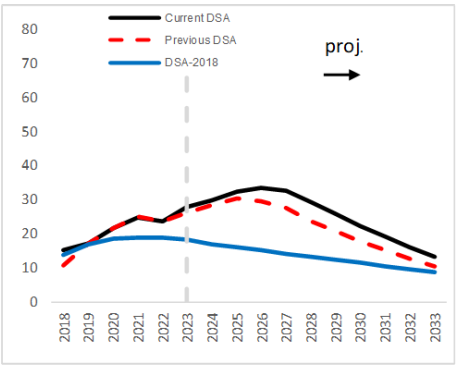
	Projections 1/										
	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033
PV of Debt-to-GDP Ratio											
Baseline	**47**	**47**	**48**	**47**	**45**	**43**	**39**	**36**	33	30	28
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	**47**	**49**	**50**	**52**	**53**	**54**	**55**	**55**	**56**	**56**	**57**
B. Bound Tests											
B1. Real GDP growth	**47**	**53**	**60**	**62**	**62**	**62**	**61**	**61**	**60**	**60**	**60**
B2. Primary balance	**47**	**51**	**57**	**57**	**54**	**52**	**49**	**45**	**42**	**39**	**37**
B3. Exports	**47**	**56**	**74**	**74**	**72**	**70**	**66**	**62**	**57**	**52**	**48**
B4. Other flows 3/	**47**	**50**	**53**	**53**	**51**	**49**	**45**	**42**	**38**	**35**	32
B5. Depreciation	**47**	**50**	**49**	**48**	**45**	**41**	**37**	33	29	25	21
B6. Combination of B1-B5	**47**	**49**	**55**	**55**	**53**	**51**	**49**	**46**	**43**	**40**	**38**
C. Tailored Tests											
C1. Combined contingent liabilities	**47**	**64**	**64**	**64**	**61**	**59**	**56**	**52**	**49**	**46**	**43**
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	**47**	**54**	**63**	**71**	**76**	**79**	**80**	**78**	**78**	**77**	**76**
C4. Market Financing	**47**	**47**	**48**	**47**	**45**	**43**	**39**	**36**	33	30	28
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio											
Baseline	259	261	259	252	231	217	195	174	154	142	131
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	259	270	274	279	276	280	273	267	261	265	270
B. Bound Tests											
B1. Real GDP growth	259	292	320	327	316	313	300	287	276	276	278
B2. Primary balance	259	282	309	302	279	264	240	217	196	184	172
B3. Exports	259	312	400	396	373	359	328	294	263	245	227
B4. Other flows 3/	259	277	289	283	261	247	223	199	177	164	151
B5. Depreciation	259	279	268	257	231	211	184	158	133	116	99
B6. Combination of B1-B5	259	274	297	294	274	262	240	219	199	188	179
C. Tailored Tests											
C1. Combined contingent liabilities	259	355	348	340	316	301	275	250	227	215	204
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	259	390	438	485	462	448	411	372	356	356	356
C4. Market Financing	259	261	259	252	231	217	195	174	154	142	131
Debt Service-to-Revenue Ratio											
Baseline	85	83	83	79	70	68	55	51	46	42	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2023-2033 2/	85	83	81	75	74	88	86	89	89	93	97
B. Bound Tests											
B1. Real GDP growth	85	90	102	107	102	105	95	95	93	95	98
B2. Primary balance	85	83	98	111	93	87	71	67	62	59	57
B3. Exports	85	83	85	86	76	74	63	63	57	52	50
B4. Other flows 3/	85	83	84	81	71	69	57	54	48	44	42
B5. Depreciation	85	79	81	78	69	70	55	51	45	41	39
B6. Combination of B1-B5	85	83	86	88	79	78	65	61	57	53	52
C. Tailored Tests											
C1. Combined contingent liabilities	85	83	150	129	106	97	82	78	74	70	67
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	85	112	115	114	118	119	100	90	83	80	83
C4. Market Financing	85	83	83	79	70	68	55	51	46	42	40

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

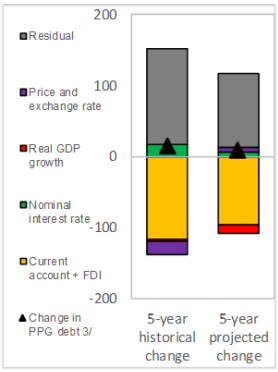
Figure 3. Papua New Guinea: Drivers of Debt Dynamics – Baseline Scenario

External debt

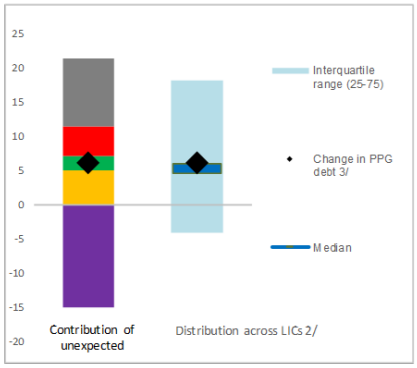
Gross Nominal PPG External Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)

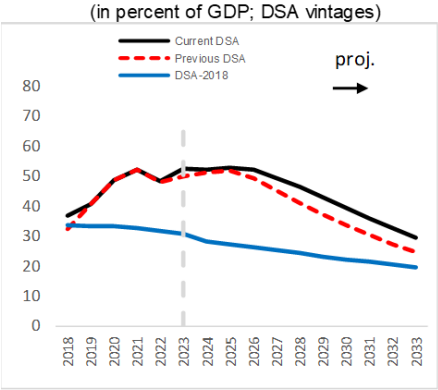


Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)

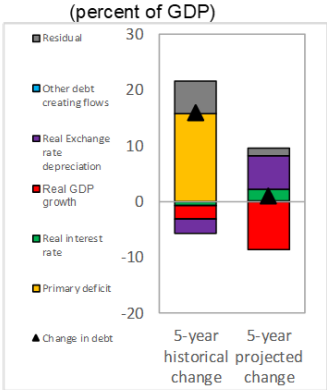


Public debt

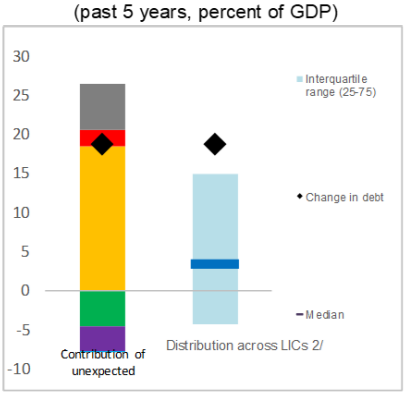
Gross Nominal Public Debt (in percent of GDP; DSA vintages)



Debt-creating flows (percent of GDP)



Unexpected Changes in Debt 1/ (past 5 years, percent of GDP)



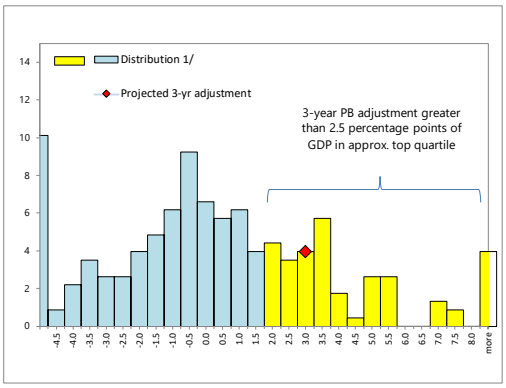
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

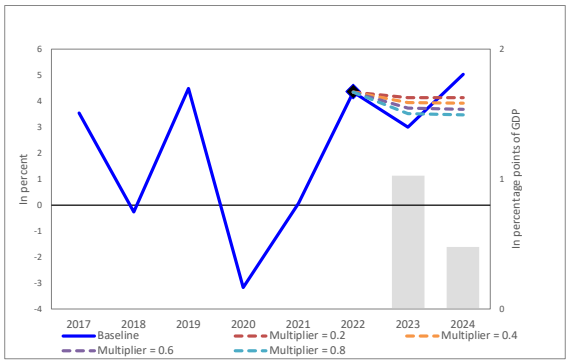
Figure 4. Papua New Guinea: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



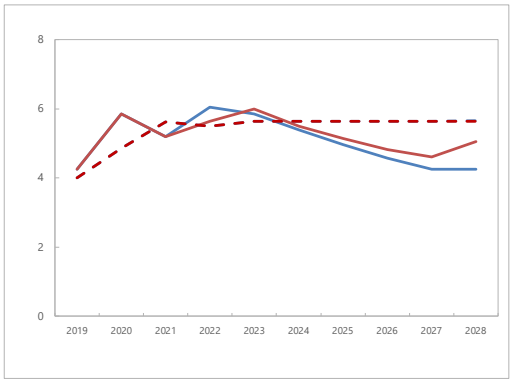
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



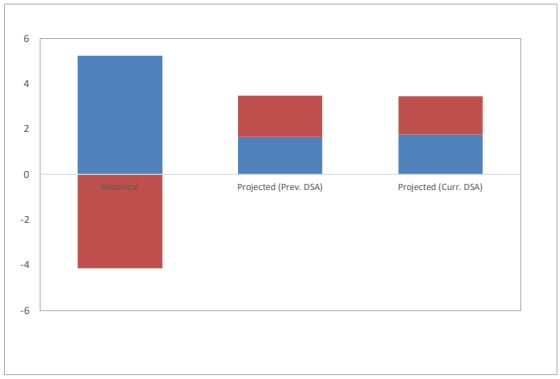
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Curr. DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Curr. DSA

Contribution to Real GDP Growth
(percent, 5-year average)

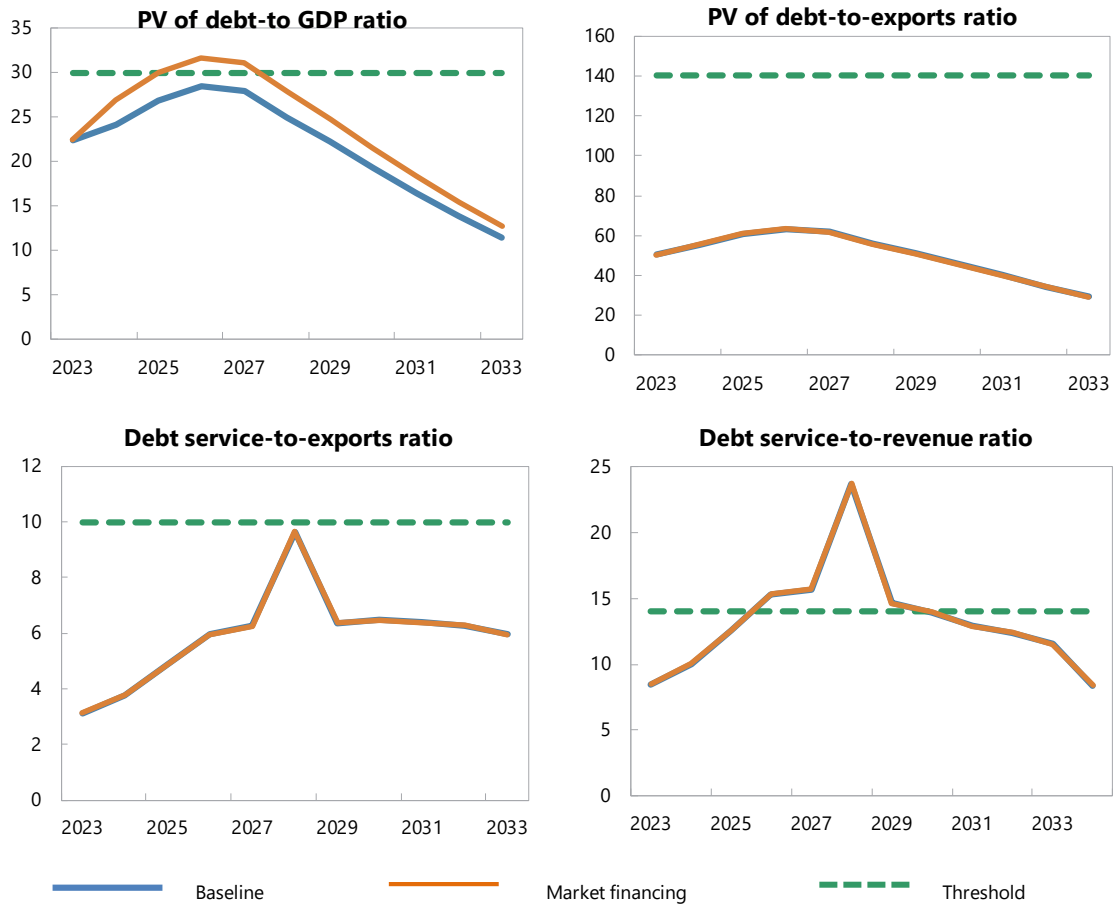


Contribution of other factors
Contribution of government capital

Figure 5. Papua New Guinea: Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	17	557
Breach of benchmark	Yes	No
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.